Final Project

Short Movie Clip Generation

1. Project Overview:

You will design, implement, and iteratively improve a text-to-video generation pipeline powered by the Diffusion model. The system takes a natural-language scene description as input and produces a short, temporally coherent video clip. You must address two core technical challenges — temporal coherence and text-video semantic alignment — and document your work in a scientific paper.

2. Inspiring Idea — Theme Selection

Your first task is to define a compelling, creative, and meaningful theme that will drive your entire project.

This is not just a prompt — it is the conceptual foundation of your system. Your chosen idea should guide:

- The design of your prompts
- The type of motion and scenes generated
- The evaluation of semantic alignment
- The storytelling in your final demo and paper

3. Technical Requirements

1- Mathematical Depth:

Your project must demonstrate a **solid mathematical understanding** of diffusion models.
The focus is not on long derivations, but on **clear explanation, intuition, and connection to your implementation**.

You should be able to explain *how math influences your model's behavior and results*.

- Diffusion Process (DDPM)

  You must explain:

1- The forward process (adding noise step by step)
2- The reverse process (denoising)

  Include:

- What happens to the image as noise increases
- How the model learns to reconstruct it

- Score Function (Basic Idea)

  Explain the idea of:

$$\nabla_x \log\, p(x)$$

  Focus on:

  - Intuition (not heavy theory)
  - Why learning noise is like learning this function

- Training Objective (ELBO → Loss)

  You should explain:

  - Why do diffusion models use a probabilistic objective
  - How it simplifies to a **mean squared error (MSE) loss**

  No full derivation required — just the key idea.

- Noise Schedule

  You must:

  - Compare at least two types (e.g., linear vs cosine)
  - Explain how they affect:
    - Training stability
    - Output quality

- Video Modeling

Explain how your model handles **time and motion**:

- Why frame-by-frame generation is problematic
- How your method improves **temporal consistency** (e.g., attention, 3D layers, etc.)

- Loss Variants

Briefly compare:

- Noise prediction (ε)
- Image prediction (x_0)
- Velocity prediction (v)

Focus on:

- When each is useful
- What your model uses

- Classifier-Free Guidance (CFG)

Explain:

- How guidance works
- Effect of guidance scale:
  - Better alignment vs less diversity

Include simple examples if possible.

- Sampling Methods

Compare:

- DDPM
- DDIM

(Optional: DPM-Solver)

Explain:

- Speed vs quality trade-off
- Which method do you use and why

Note: What We Expect

- Clear explanations (not copied text)
- Simple equations were needed
- Direct connection to:
  - Your model
  - Your results

2- Architecture Deep Dive

You must fully document the architecture of your chosen model, including:

- 3D U-Net structure: spatial downsampling, temporal downsampling, residual blocks
- Attention mechanisms: spatial self-attention, temporal self-attention, cross-attention for text conditioning
- Position encodings for temporal dimension (sinusoidal, rotary, or learned)
- Conditioning mechanisms: text encoder (CLIP / T5), frame-level conditioning tokens
- Latent vs. pixel-space diffusion; VAE encoder/decoder role in latent video models

3- Loss Functions — Detailed Analysis

Provide the full mathematical formulation of each loss variant, then justify which variant your model uses and why.

4- Weakness analysis and Enhancement

This is the highest component of your project. Every team must formally identify and then address at least two weaknesses

5- Evaluation Metrics

Metric	What It Measures	Tool / Library
FVD (Fréchet Video Distance)	Overall quality + realism of video distribution	styleganvideo / torch-fidelity
IS (Inception Score)	Sharpness and diversity of frames	torchmetrics

CLIP-SIM	Semantic alignment between prompt and video	openai/clip-vit
SSIM / PSNR	Frame-level pixel reconstruction quality	skimage.metrics
LPIPS (temporal)	Perceptual frame difference over time (flickering)	lpips library
Flow Warping Error	Optical-flow-based temporal consistency	RAFT + custom metric
User Study Score	Subjective realism, coherence, creativity (1–5 scale)	Custom survey (≥10 raters)

6- Scientific Paper Structure

Your paper must follow the IEEE double-column format and include all the following sections:

Section	Required Content
Abstract	Problem, method, key results, takeaway in 150–250 words
1. Introduction	Motivation, problem statement, your theme, contributions list
2. Background & Math	Clear mathematical explanation of DDPM, loss functions, and architecture (no full derivations required)
3. Model Architecture	Chosen model architecture, 3D U-Net, attention, conditioning — with diagrams
4. Loss Function Analysis	All loss variants with formulas; justify your model's choice; your modifications
5. Weakness Analysis	Formal experiments, quantitative evidence, root-cause hypothesis
6. Proposed Enhancements	Mathematical formulation of each enhancement; implementation details
7. Experiments & Results	Metrics table, ablation study, qualitative samples, user study, if any
8. Discussion	Interpretation of results, limitations, and future work
9. Conclusion	Summary of contributions and findings
References	All citations in IEEE format

7- Deliverables and Timeline

Phase 1 — Week 13: Foundation & Analysis

✓ Selected theme + motivation

✓ Selection of the Pretrained Model Architecture

✓ Clear mathematical explanation of DDPM, loss functions, and architecture (no full derivations required)

✓ Weakness analysis: formal experiments quantifying ≥2 identified weaknesses with metrics

✓ Paper draft: Sections 1 (Introduction), 2 (Background & Math), 3 (Model Architecture), 4 (Weaknesses)

Phase 2 — Week 15: Enhancement & Final Paper

✓ Fully implemented enhancement(s) with mathematical motivation

✓ Ablation study comparing: baseline vs. enhancement A vs. enhancement B vs. combined

✓ Final generated video (≥4 seconds) demonstrating improvement

✓ Complete scientific paper: all sections finalized

✓ Code repository: clean, commented, with requirements.txt and generation script

Bonus: Text – to – audio

As an optional high-level enhancement, students may extend the system by adding a Text-to-Audio (TTS) module that generates synchronized narration or sound for the generated video.

This feature must go beyond basic speech synthesis and demonstrate advanced system design and AI integration.

To increase difficulty, the implementation must include at least two of the following:

- Multi-voice support (different speakers or styles)
- Emotion-controlled speech generation (happy, sad, neutral, etc.)
- Adjustable speech parameters (speed, pitch, emphasis)
- Context-aware synthesis (proper pauses, punctuation handling, prosody)
- Neural TTS model integration (not basic libraries)
- Streaming or chunk-based generation for long scripts

Bonus Evaluation Criteria

This component will significantly enhance the final grade if:

- It is properly integrated with the video pipeline
- It improves the storytelling quality of the generated clips
- It demonstrates technical depth beyond standard TTS usage